

HORNSBY GIRLS HIGH SCHOOL



Mathematics Extension 1

Year 11

Assessment 2 Term 2 2022

STUDENT NUMBER: _____

General Instructions

- Reading Time – 5 minutes
- Working Time – 60 minutes
- Write using black or blue pen
Black pen is preferred
- NESA-approved calculators may be used
- In Questions 6 – 9, show relevant mathematical reasoning and/or calculations
- Marks may be deducted for untidy and poorly arranged work
- Do not use correction fluid or tape
- Do not remove this paper from the examination

Total marks – 44

Section I Pages 3 – 4

5 marks

Attempt Questions 1 – 5

Answer on the Objective Response Answer Sheet provided

Section II Pages 5 – 8

39 marks

Attempt Questions 6 – 9.

Question	1-5	6	7	8	9	Total
Total	/5	/10	/11	/8	/10	/44

This assessment task constitutes 30% of the Preliminary Higher School Certificate Course School Assessment

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Section I

5 marks

Attempt Questions 1 – 5

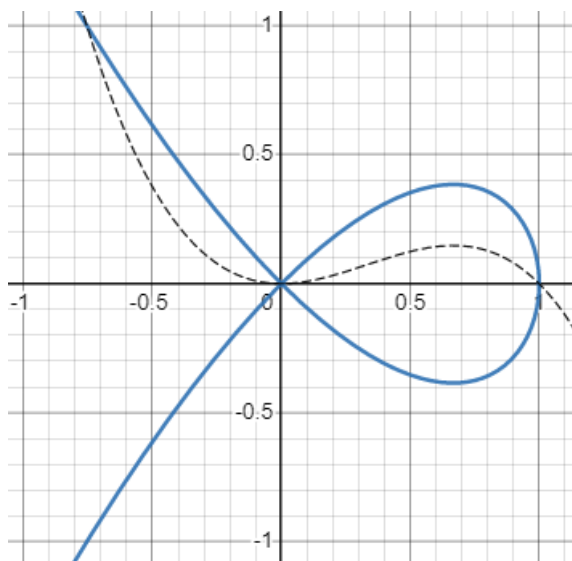
Allow about 10 minutes for this section

Use the Objective Response answer sheet for Questions 1 – 5

Q1. The range of $f(g(x))$ if $f(x) = x + 4$ and $g(x) = x^2$ is

- (A) $[0, \infty)$
- (B) $(-\infty, \infty)$
- (C) $(4, \infty)$
- (D) $[4, \infty)$

Q2. The dotted curve represents $y = f(x)$.



Which of the following expressions best represents the solid curve?

- (A) $y = |f(x)|$
- (B) $y = \sqrt{f(x)}$
- (C) $y^2 = f(x)$
- (D) $y = \frac{1}{f(x)}$

Q3. If $f(x) = 3x^2 - 1$ for $x \geq 0$, then the domain of $f^{-1}(x)$ is

- (A) $x \geq 1$
- (B) $y \geq 0$
- (C) $x \geq -1$
- (D) $x \geq 0$

Q4. Which of the following is **NOT** a possible pair of parametric equations for the Cartesian equation $y = 2x + 4$?

- (A) $x = t, y = 2t + 4$
- (B) $x = t + 2, y = 2t$
- (C) $x = \frac{t}{2}, y = t + 4$
- (D) $x = 2t, y = 4t + 4$

Q5. If $f(x+1) = 2x^2 + x - 2$, find $f(x)$.

- (A) $f(x) = 2x^2 - 3x + 1$
- (B) $f(x) = 2x^2 - 3x - 4$
- (C) $f(x) = 2x^2 + x - 3$
- (D) $f(x) = 2x^2 - 3x - 1$

End of Section I

Section II

30 marks

Attempt Questions 6-8

Allow about 50 minutes for this section

Question 6 (10 marks)

Start a new answer booklet

(a) Simplify $\frac{3ab^2}{x-2y} \div \frac{12ab-6a}{x^3-8y^3}$ 2

(b) Find the values of x and y if $\frac{5\sqrt{3}+1}{2-\sqrt{3}} = x + y\sqrt{3}$. 2

(c) Simplify $12^{2n} \div 3^n \times 48^{1-n}$ 3

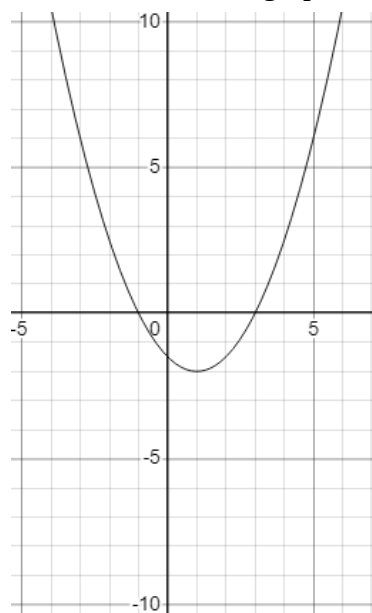
(d) Use the method of completing the square to find the exact values of x if $3x^2 + 6x - 2 = 0$. 3

Question 7 (11 marks) **Start a new answer booklet**

(a) Solve $\frac{x-2}{x-1} \leq 2$. **3**

(b) Solve $|2x+1| + |x-2| \leq 3$ **3**

(c) The diagram below is a sketch of the graph of the function $f(x) = \frac{1}{2}(x+1)(x-3)$



Use the graphs provided on the separate answer sheet, to graph the following transformations showing all asymptotes and intercepts.

(i) $y = f(|x|)$ **1**

(ii) $y = [f(x)]^2$ **2**

(iii) $y = \frac{1}{f(x)}$ **2**

Question 8 (8 marks) **Start a new answer booklet**

(a) Solve $\cos \frac{\theta}{2} = \frac{-\sqrt{3}}{2}$ for $-360^\circ \leq \theta \leq 360^\circ$ **2**

(b) Prove that $\left(\sin \theta + \frac{\cos^2 \theta}{\sin \theta} \right)^2 = 1 + \cot^2 \theta$ **3**

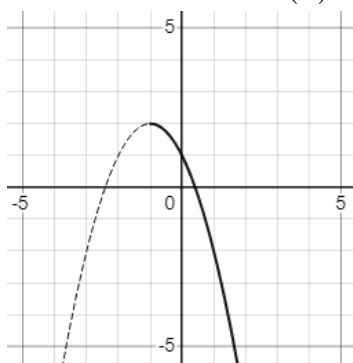
(c) Solve $2 \cos^2 x + \sin x + 1 = 0$ for $x \in [0, 2\pi]$. **3**

Question 9 (10 marks)**Start a new answer booklet**

- (a) Find the Cartesian equation for the parametric equations $x = 3\sin \theta$, $y = 3\cos \theta$, $0 \leq \theta \leq \pi$ and state the domain of the Cartesian equation.

3

- (b) Consider the curve $f(x) = 2 - (x+1)^2$ and its graph where $x \geq -1$



- (i) The largest possible domain involving positive values of x for which $f(x)$ has an inverse function $f^{-1}(x)$ is $x \geq -1$. Explain how this restricted domain for the function $f(x)$ allows the inverse $f^{-1}(x)$ to be a function.

1

- (ii) State the domain of $y = f^{-1}(x)$.

1

- (iii) Find the equation of $y = f^{-1}(x)$.

3

- (iv) For the restricted domain, sketch the inverse function showing the main features.

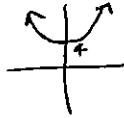
2**End of paper**

Year 11 Maths Ex+1

Task 2, 2022

1) $f(x) = x + 4$, $g(x) = x^2$

$f(g(x)) = x^2 + 4$

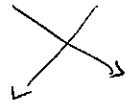
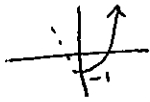


R: $y > 4$

$[4, \infty)$ (D)

2) (C)

3) $f: y = 3x^2 - 1$ D: $x \geq 0$, R: $y \geq -1$



$f^{-1}: x = 3y^2 - 1$ D: $x \geq -1$ R: $y \geq 0$

(C)

4) A. $x = t$, $y = 2t + 4$
 $y = 2x + 4$ ✓

B. $x = t + 2$, $y = 2t$
 $t = x - 2$ $y = 2(x - 2)$
 $= 2x - 4$ ✗ (B)

C. $x = \frac{t}{2}$, $y = t + 4$
 $t = 2x$ $y = 2x + 4$ ✓

D. $x = 2t$, $y = 4t + 4$
 $t = \frac{x}{2}$ $y = 4(\frac{x}{2}) + 4$
 $= 2x + 4$ ✓

5) If $f(x+1) = 2x^2 + x - 2$

then $f((x-1)+1) = 2(x-1)^2 + (x-1) - 2$
 $= 2(x^2 - 2x + 1) + x - 1 - 2$
 $= 2x^2 - 4x + 2 + x - 3$
 $= 2x^2 - 3x - 1$ (D)

or $f(x+1) = 2(x^2 + 2x + 1) - 3x - 4$

$f(x+1) = 2(x+1)^2 - 3(x+1) - 1$

$\therefore f(x) = 2x^2 - 3x - 1$

$$\begin{aligned}
 \text{b) a) } & \frac{3ab^2}{x-2y} \div \frac{12ab-6a}{x^3-8y^3} \\
 & = \frac{3ab^2}{x-2y} \times \frac{(x-2y)(x^2+2xy+4y^2)}{6a(2b-1)} \\
 & = \frac{b^2(x^2+2xy+4y^2)}{2(2b-1)}
 \end{aligned}$$

Markers feedback

1 for factorising difference of 2 cubes
1 for x reciprocal and simplifying

Poorly done. Many students didn't factorise the cubic expression correctly.

$$\begin{aligned}
 \text{b) } & \frac{5\sqrt{3}+1}{2-\sqrt{3}} = \frac{5\sqrt{3}+1}{2-\sqrt{3}} \times \frac{2+\sqrt{3}}{2+\sqrt{3}} \\
 & = \frac{10\sqrt{3}+5 \times 3+2+\sqrt{3}}{4-3} \\
 & = \frac{11\sqrt{3}+17}{1} \\
 & = 17+11\sqrt{3} \\
 & = x+y\sqrt{3} \\
 & \text{where } x=17, y=11
 \end{aligned}$$

1 for multiplying top and bottom by conjugate of denominator and expanding

1 for simplifying and equating

Generally done well.

However some students brought the x & y terms into the rationalisation of the denominator

$$\begin{aligned}
 \text{c) } & 12^{2n} \div 3^n \times 48^{1-n} \\
 & = (3 \times 2^2)^{2n} \times 3^{-n} \times (2 \times 3 \times 2^3)^{1-n} \\
 & = 3^{2n} \times 2^{4n} \times 3^{-n} \times 2^{4-4n} \times 3^{1-n} \\
 & = 3^{2n-n+1-n} \times 2^{4n+4-4n} \\
 & = 3 \times 2^4 \\
 & = 3 \times 16 \\
 & = 48
 \end{aligned}$$

1 for expressing with bases 2 and 3

1 for correct indices

1 for simplifying

Very poorly done! Many students did not use the rule

$$\frac{a^n}{b^n} = \left(\frac{a}{b}\right)^n$$

$$\begin{aligned}
 \text{d) } & 3x^2+6x-2=0 \\
 & 3x^2+6x = 2 \\
 & 3(x^2+2x) = 2 \\
 & 3(x^2+2x+1) = 2+3 \\
 & 3(x+1)^2 = 5 \\
 & (x+1)^2 = \frac{5}{3} \\
 & x+1 = \pm \sqrt{\frac{5}{3}} \\
 & x = -1 \pm \sqrt{\frac{5}{3}}
 \end{aligned}$$

1 for completing the square

1.

1.

Generally done well!

7) a) $\frac{x-2}{x-1} \leq 2$

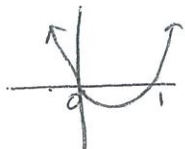
$$\frac{x-2}{x-1} \times (x-1)^2 \leq 2(x-1)^2 \quad |.$$

$$x^2 - x - 2x + 2 \leq 2(x^2 - 2x + 1)$$

$$x^2 - 3x + 2 \leq 2x^2 - 4x + 2$$

$$0 \leq x^2 - x$$

$$x(x-1) \geq 0$$



$$x \leq 0, x \geq 1 \quad |.$$

$$\text{but } x \neq 1$$

$$x \leq 0, x \geq 1 \quad |.$$

OR

if $x > 1$

then $x-2 \leq 2(x-1)$

$$x-2 \leq 2x-2$$

$$0 \leq x$$

$$x > 0 \quad |.$$

but $x > 1$

$$\therefore x > 1 \quad |.$$

if $x < 1$

then $x-2 > 2(x-1)$

$$x-2 > 2x-2$$

$$0 > x$$

$$x \leq 0$$

which satisfies $x < 1$

$$\therefore x \leq 0 \quad |.$$

$$x \leq 0, x > 1$$

generally well done
but a lot of
students missed
that $x \neq 1$

Those who didn't
graph their quadratic
inequality found
difficulty.

Those who tried to
use this method
generally missed
the respective domains
that affected their
answer.

b) $|2x+1| + |x-2| \leq 3$



if $x \leq -\frac{1}{2}$

$$-(2x+1) - (x-2) \leq 3$$

$$-2x-1-x+2 \leq 3$$

$$-3x+1 \leq 3$$

$$-3x \leq 2$$

$$x \geq -\frac{2}{3}$$

but $x \leq -\frac{1}{2}$

$$\therefore -\frac{2}{3} \leq x \leq -\frac{1}{2}$$

if $-\frac{1}{2} < x < 2$

$$2x+1 - (x-2) \leq 3$$

$$2x+1-x+2 \leq 3$$

$$x+3 \leq 3$$

$$x \leq 0$$

but $-\frac{1}{2} < x < 2$

$$\therefore -\frac{1}{2} < x \leq 0$$

if $x \geq 2$

$$2x+1 + x-2 \leq 3$$

$$3x-1 \leq 3$$

$$3x \leq 4$$

$$x \leq \frac{4}{3}$$

but $x \geq 2$

$\therefore$ no solution

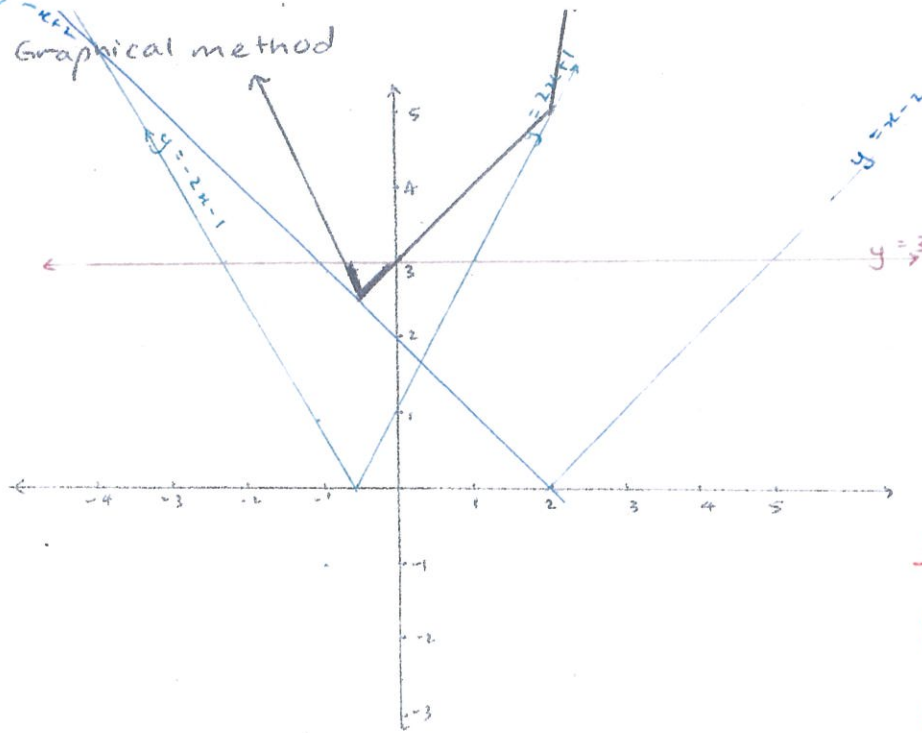
$$-\frac{2}{3} \leq x \leq 0$$

If using Algebra.

Poorly done question
in general, mainly
again due to
not taking into
account the domains
of each part
properly

OR ---

7 b) Graphical method



$$-2x-1 + -x+2 = 3$$

$$-3x+1 = 3$$

$$-3x = 2$$

$$x = -\frac{2}{3}$$

$$2x+1 + -x+2 = 3$$

$$x+3 = 3$$

$$x = 0$$

$$-\frac{2}{3} \leq x \leq 0$$

• When graphing there is a need

to make it large enough to work effectively with and

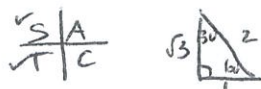
• Labelling the branches helps when doing the sum of the absolutes correctly.

8) a)

$$\cos \frac{\theta}{2} = -\frac{\sqrt{3}}{2}$$

$$\text{for } -360^\circ \leq \theta \leq 360^\circ$$

$$-180^\circ \leq \frac{\theta}{2} \leq 180^\circ$$



related angle is 30°

$$\frac{\theta}{2} = -180^\circ + 30^\circ, 180^\circ - 30^\circ$$

$$\frac{\theta}{2} = -150^\circ, 150^\circ$$

$$\theta = -300^\circ, 300^\circ$$

Markers feedback

some students
didn't use degree

b) Prove that $\left(\sin \theta + \frac{\cos^2 \theta}{\sin \theta}\right)^2 = 1 + \cot^2 \theta$

$$\text{LHS} = \left(\sin \theta + \frac{\cos^2 \theta}{\sin \theta}\right)^2$$

$$= \sin^2 \theta + 2\sin \theta \times \frac{\cos^2 \theta}{\sin \theta} + \frac{\cos^4 \theta}{\sin^2 \theta}$$

$$= \sin^2 \theta + 2\cos^2 \theta + \frac{\cos^4 \theta}{\sin^2 \theta}$$

$$= \sin^2 \theta + \cos^2 \theta + \cos^2 \theta + \frac{\cos^2 \theta \cdot \cos^2 \theta}{\sin^2 \theta}$$

$$= 1 + \cos^2 \theta \left(1 + \frac{\cos^2 \theta}{\sin^2 \theta}\right) \quad (\sin^2 \theta + \cos^2 \theta = 1)$$

$$= 1 + \cos^2 \theta (1 + \cot^2 \theta)$$

$$= 1 + \cos^2 \theta \times \operatorname{cosec}^2 \theta \quad (1 + \cot^2 \theta = \operatorname{cosec}^2 \theta)$$

$$= 1 + \cos^2 \theta \times \frac{1}{\sin^2 \theta}$$

$$= 1 + \frac{\cos^2 \theta}{\sin^2 \theta}$$

$$= 1 + \cot^2 \theta$$

$$\text{LHS} = \text{RHS}$$

or

$$\text{LHS} = \sin^2 \theta + 2\cos^2 \theta + \frac{\cos^4 \theta}{\sin^2 \theta}$$

$$= \frac{\sin^4 \theta + 2\sin^2 \theta \cos^2 \theta + \cos^4 \theta}{\sin^2 \theta}$$

$$= \frac{(\sin^2 \theta + \cos^2 \theta)^2}{\sin^2 \theta} \quad (\sin^2 \theta + \cos^2 \theta = 1)$$

$$= \frac{1^2}{\sin^2 \theta}$$

$$= \operatorname{cosec}^2 \theta$$

$$= 1 + \cot^2 \theta$$

$$\text{LHS} = \text{RHS}$$

or:

$$\text{RHS} = \left(\frac{\sin^2 \theta + \cos^2 \theta}{\sin \theta}\right)^2$$

$$= \left(\frac{1}{\sin \theta}\right)^2$$

$$= \operatorname{cosec}^2 \theta$$

$$= 1 + \cot^2 \theta$$

mostly well done.
some students
need to be more
familiar with
Trig identities.

$$8) c) 2\cos^2 x + \sin x + 1 = 0$$

$$2(1 - \sin^2 x) + \sin x + 1 = 0$$

$$2 - 2\sin^2 x + \sin x + 1 = 0$$

$$-2\sin^2 x + \sin x + 3 = 0$$

$$2\sin^2 x - \sin x - 3 = 0$$

1. for quadratic
in terms of $\sin x$

(1)

$$(2\sin x - 3)(\sin x + 1) = 0$$

$$2\sin x = 3$$

$$\text{or } \sin x = -1$$

$$\sin x = \frac{3}{2}$$

$$x = \frac{3\pi}{2}$$

(1)

(1)

$$\text{but } -1 \leq \sin x \leq 1$$

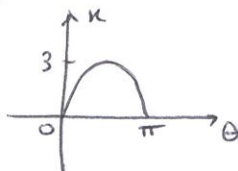
$\therefore$ no solution

← need to justify
why this solution is rejected

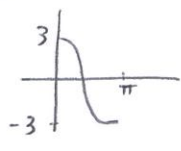
$$\therefore x = \frac{3\pi}{2}$$

some students use degree instead of radian

9) a) $x = 3\sin\theta$, $y = 3\cos\theta$, $0 \leq \theta \leq \pi$



$\therefore 0 \leq x \leq 3$



$-3 \leq y \leq 3$

$\sin\theta = \frac{x}{3}$

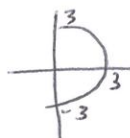
$\cos\theta = \frac{y}{3}$

but $\sin^2\theta + \cos^2\theta = 1$

$\left(\frac{x}{3}\right)^2 + \left(\frac{y}{3}\right)^2 = 1$

$\frac{x^2}{9} + \frac{y^2}{9} = 1$

$x^2 + y^2 = 9$ for $0 \leq x \leq 3$



Most students were able to find the Cartesian equation $\frac{x^2}{9} + \frac{y^2}{9} = 1$ but a surprisingly large number then made errors after that e.g. $x^2 + y^2 \neq 1$

very few students (approx 10) were able to find the domain correctly. Most incorrectly stated $-3 \leq x \leq 3$.

Horizontal/vertical line test is a visual technique that we may use but it is not a reason and should not be given as the only explanation.

b) (i) The restricted domain makes $y = f(x)$ a one-to-one function so its inverse is also a one-to-one function.

or

The restricted domain makes every y value correspond to only one x value so that the inverse has only one y value for every x value, hence making the inverse a function.

(ii) Domain of $f(x)$ is $x \geq -1$
Range of $f(x)$ is $y \leq 2$

$\therefore$ Domain of $f^{-1}(x)$ is $x \leq 2$

(iii) $f: y = 2 - (x+1)^2$

$f^{-1}: x = 2 - (y+1)^2$

$x - 2 = -(y+1)^2$

$2 - x = (y+1)^2$

$y+1 = \pm\sqrt{2-x}$

$y = -1 \pm \sqrt{2-x}$ but D: $x \leq 2$

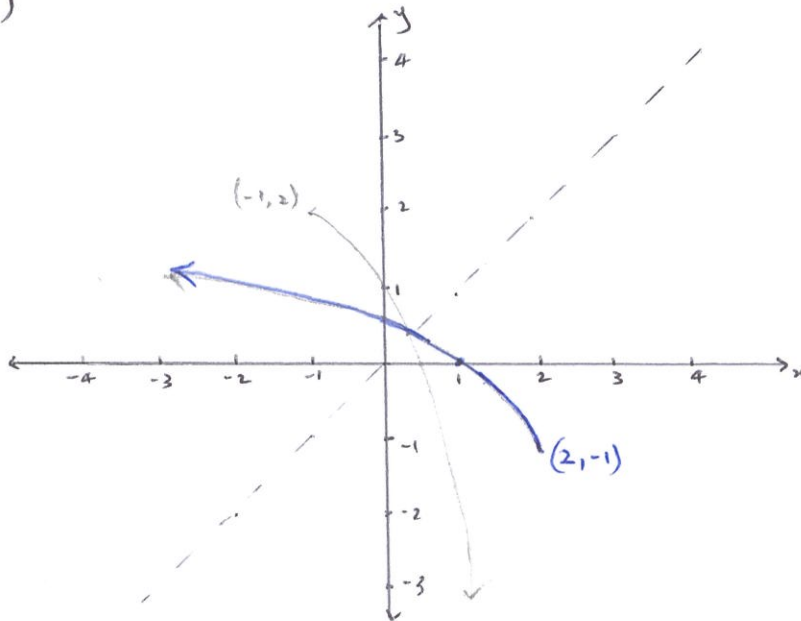
R: $y \geq -1$

$\therefore y = -1 + \sqrt{2-x}$

Most students were able to find $y = -1 \pm \sqrt{2-x}$ but then many stated $x \geq 2$ instead of $y \geq -1$ to justify why the negative case is discarded to give $y = -1 + \sqrt{2-x}$

a) b)

(iv)



1 for passing through $(2, -1)$, $(1, 0)$
and basic shape

1 for accuracy

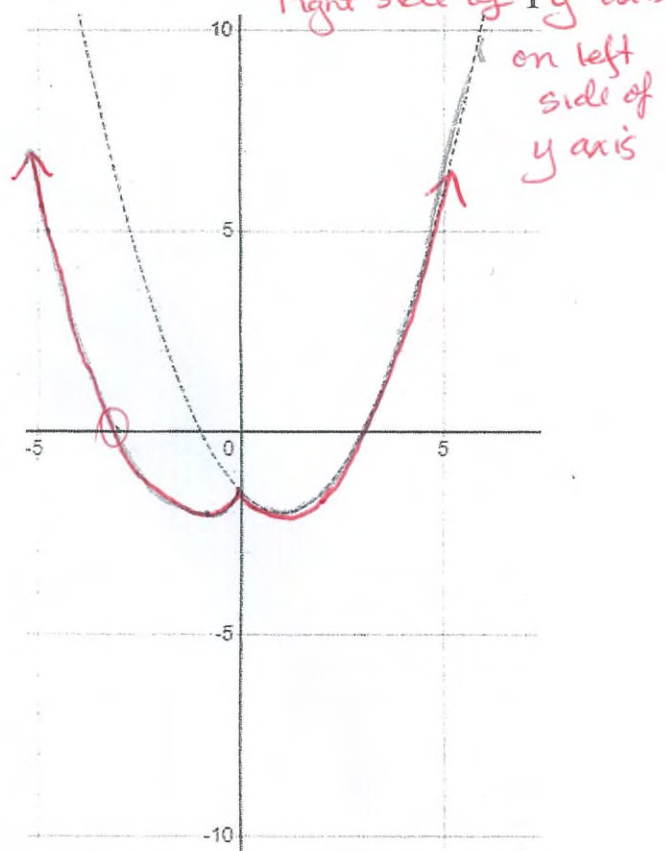
Question 7

The diagrams below are graphs of the function $f(x) = \frac{1}{2}(x+1)(x-3)$.

Use the graphs provided to sketch each of the following, showing all asymptotes and intercepts.

(i) $y = f(|x|)$

*-x values match +x values
i.e mirror image of right side of y axis*

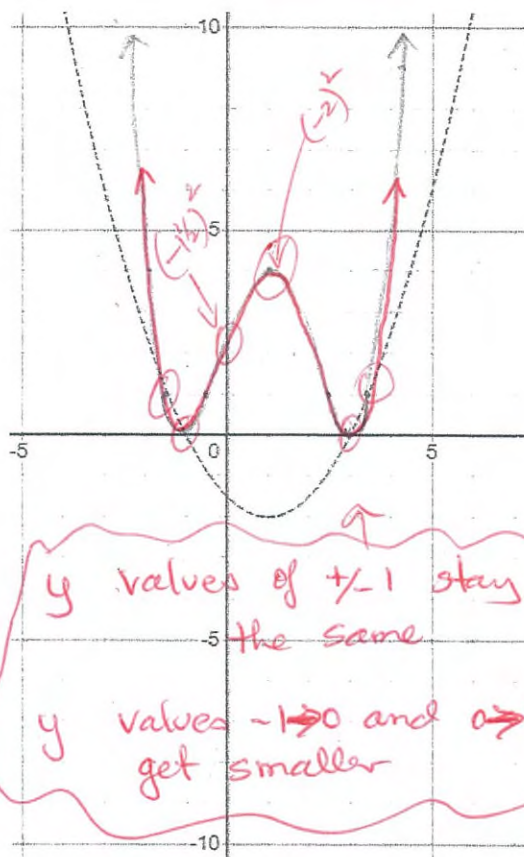


These questions produced a mixed bag of results mainly because of

1. Recognition of what was asked was not known
2. Inaccuracy and not going through basic points to produce an accurate transformation

(ii) $y = [f(x)]^2$

y values are squared



(iii) $y = \frac{1}{f(x)^2}$

y values are reciprocated

